PES UNIVERSITY

Department of Computer Science and Engineering

SOFTWARE REQUIREMENTS SPECIFICATION

Electronics Store Inventory Management System

TechZone Electronics | Group 9

Document Item	Details
Version	1.0
Prepared For	Software Engineering – Level 3 Mini-Project
Case Study	TechZone Electronics (Fictional Electronics Retail Store)
Development Model	Agile / Incremental Sprints
Prepared By	Team 9
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Team Members

Sl. No.	USN	Student Name	Primary Functional Ownership
1	PES1UG24CS068	Annamalai N	User Authentication & Role Management
2	PES1UG24CS077	Apurv Kumar Singh	Product & Inventory Management
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1. Introduction and Purpose

1.1 Purpose

This Software Requirements Specification (SRS) defines the requirements for the Electronics Store Inventory Management System proposed for TechZone Electronics, a fictional small-to-medium electronics retail shop used as the project case study. The system is a web-based internal application for managing products, stock, suppliers, purchases, sales, user access, and operational reports.

The SRS provides a common reference for the development team, faculty/evaluators, testers, and future maintainers. It converts project objectives into clear, uniquely identified, measurable, and testable requirements.

1.2 Intended Audience and Reading Suggestions

- Project team members – requirements, ownership, design and implementation basis.
- Faculty/evaluators – completeness, feasibility, clarity and traceability assessment.
- Developers – architecture, API, database and UI implementation.
- Testers – derivation of unit, integration, system and acceptance tests.
- Maintainers – understanding intended behavior and business rules.

Sections 1–7 provide context; Sections 8–10 define interfaces, analysis models and functional requirements; Sections 11–13 define quality, security and business rules; Sections 14–16 define data, terminology and traceability.

1.3 Requirement Conventions

- FR-xxx = functional requirement; NFR-xxx = non-functional requirement; SEC-xxx = security requirement; BR-xxx = business/domain rule; UC-xx = use case.
- “Shall” denotes mandatory behavior.
- Planned architecture, code and test references in the RTM are placeholders to be updated after implementation.

1.4 References

Reference	Description
PES University SE Mini-Project Guidelines, 2026	Faculty-provided project and SRS guidance/template.
Team 9 Synopsis – Electronics Store Inventory Management System	Project proposal defining problem, scope, users, features, ownership, plan and technology stack.
React / Django REST Framework / PostgreSQL documentation	Technical references for the planned implementation stack.

2. Product Scope

The system will centralize internal inventory and transaction operations of TechZone Electronics. It will maintain consistent product and stock records while connecting purchasing and sales events to inventory changes.

2.1 In-Scope

- Authentication, user management and role-based access.
- Product/category management and product search.
- Inventory quantity tracking, adjustments, low-stock identification and inventory history.
- Supplier management and purchase order management.
- Recording received purchases and increasing stock accordingly.

- Sales transactions and inventory reduction.
- Operational reports for inventory, purchases, sales and low-stock items.
- Validation, filtering and common error handling.

2.2 Out-of-Scope

- Public e-commerce storefront and online ordering.
- Online payment gateway integration.
- Delivery/courier management.
- Full accounting, payroll, taxation or ERP functions.
- AI demand forecasting.
- Multi-store enterprise management.
- Customer loyalty/CRM and marketing automation.
- Supplier payment settlement or banking integration.

3. Intended Users

User / Stakeholder	Primary Need	Typical Activities
Store Administrator	Control access and master data	Users/roles, products, suppliers, inventory controls, reports
Inventory / Store Staff	Maintain accurate stock	Product search, stock status, permitted adjustments, low-stock review
Purchase Staff	Manage procurement	Suppliers, purchase orders, receipts, purchase history
Sales Staff	Record retail sales	Product availability, sale entry, transaction history
Store Manager / Owner	Monitor operations	Inventory, sales, purchases and low-stock reports
Developers / Testers	Build and validate	Implementation, testing, reviews and documentation

4. Product Functions

ID	Major Function	Summary	Primary Owner
F-01	Authentication & Roles	Login, logout, users, roles and permissions.	Annamalai N
F-02	Product & Category	Product/category creation, update, search and maintenance.	Apurv Kumar Singh
F-03	Inventory	Stock quantities, adjustments, low-stock status and history.	Apurv Kumar Singh
F-04	Supplier Management	Supplier profile and status management.	Atharv Mittal
F-05	Purchase Management	Purchase orders, receipts and stock increases.	Atharv Mittal
F-06	Sales Management	Sales transactions, validation and stock decreases.	Avaneesh H D
F-07	Reporting	Inventory, purchase, sales and low-stock reports.	Avaneesh H D
F-08	Search & Filtering	Search/filter products, suppliers and transactions.	Shared

5. User Classes

5.1 Administrator

Highest-privilege operational user. Manages users/roles and has broad access needed to administer store data.

5.2 Inventory Staff

Maintains product and stock information, views availability, performs permitted adjustments and monitors low stock.

5.3 Purchase Staff

Maintains suppliers, creates purchase records and records receipt of goods.

5.4 Sales Staff

Records sales and needs access to product availability and transaction history.

5.5 Manager / Owner

Primarily reviews operational reports, inventory status, purchases and sales.

6. Operating Environment

Component	Planned Environment
Frontend	React with JavaScript; responsive browser-based UI
Backend	Python with Django REST Framework
Database	PostgreSQL
Client	Current stable Chrome/Chromium, Firefox or Edge on desktop/laptop
Server	Linux-based development/deployment environment
Tools	Git/GitHub, Jira, Docker, Jenkins, SonarQube
Testing	pytest plus API/integration/system testing
Network	LAN/Wi-Fi for development; HTTPS-enabled deployment for production

The application is intended to operate using standard web-browser and server infrastructure without specialized client hardware.

7. Constraints and Assumptions

7.1 Design and Implementation Constraints

- React/JavaScript, Django REST Framework/Python and PostgreSQL shall be used as the primary stack.
- Client-server web architecture shall be used.
- Git/GitHub and Jira shall be used for source control and project tracking.
- Role authorization shall be enforced at the backend/API level.
- Database referential integrity shall be maintained.
- The project is an academic mini-project; integrations outside the stated scope are not required.

7.2 Assumptions and Dependencies

- Authorized staff have valid credentials and a supported browser.
- Initial master data is entered by authorized staff.
- Stock is maintained in integer units.
- A sale cannot be completed when stock is insufficient.
- Received purchases increase inventory only after successful receipt recording.
- PostgreSQL, backend API and required deployment services are available.
- Approved requirement changes will be reflected through SRS revision control.

8. External Interface Requirements

8.1 User Interfaces

- Login screen with credential validation and session/logout behavior.
- Role-appropriate dashboard and navigation.
- Product/inventory tables and forms with search, filters and validation.
- Supplier/purchase screens with supplier selection, item entry, totals, status and receipt actions.
- Sales screen with product search, quantity entry, stock validation, totals and confirmation.
- Reports screen with filters and tabular summaries.
- Consistent labels, required-field indicators, success messages and error messages.
- Unauthorized actions shall be rejected by the backend even if a client request is manually constructed.

8.2 Software Interfaces

Interface	Direction	Purpose / Data
React ↔ Django REST API	Bidirectional	JSON requests/responses for application modules.
Django ↔ PostgreSQL	Bidirectional	Persistent application data through the database layer/ORM.
Git/GitHub	Outbound	Source control and collaboration.
Jenkins	Inbound/Outbound	Automated build/test pipeline.
SonarQube	Outbound	Static analysis and quality checks.

8.3 Communications Interfaces

- Frontend/backend communication shall use HTTP/HTTPS REST APIs.
- JSON shall be the primary data format.
- Production communication shall use HTTPS.
- API responses shall use appropriate HTTP status codes.
- Database connections shall remain internal and shall not be exposed directly to browsers.

9. Use Case / Analysis Models

9.1 Use Case Catalogue

ID	Use Case	Primary Actor	Goal
UC-01	Authenticate User	All users	Securely access the system according to role.
UC-02	Manage Users and Roles	Administrator	Create/update/deactivate users and assign roles.
UC-03	Manage Products	Admin / Inventory	Maintain product and category data.

ID	Use Case	Primary Actor	Goal
UC-04	Manage Inventory	Inventory Staff	View, adjust and monitor stock.
UC-05	Manage Suppliers	Purchase Staff / Admin	Maintain supplier records.
UC-06	Create Purchase Order	Purchase Staff	Record procurement from a supplier.
UC-07	Receive Purchase	Purchase Staff	Record received items and increase stock.
UC-08	Record Sale	Sales Staff	Record a sale and decrease stock.
UC-09	View Reports	Manager / Admin	Review inventory, purchase, sales and low-stock data.
UC-10	Search and Filter	Authorized users	Locate relevant records efficiently.

9.2 Main Business Flow

Authenticate User → Product/Inventory Setup → Supplier & Purchase Processing → Inventory Increase on Receipt → Sales Processing → Inventory Decrease on Sale → Reports and Low-Stock Monitoring.

9.3 Use Case Details

UC-01 Authenticate User

Main flow: Enter valid credentials → validate → establish authenticated session/token → show role-specific dashboard.

Exception handling: Invalid credentials or inactive account → deny access.

UC-02 Manage Users and Roles

Main flow: Admin opens user management → enters data → assigns role → validates → saves.

Exception handling: Duplicate account data or invalid role → reject.

UC-03 Manage Products

Main flow: Open products → create/update category/product → validate → save.

Exception handling: Duplicate SKU or invalid values → reject.

UC-04 Manage Inventory

Main flow: View stock → enter permitted adjustment and reason → validate → save transaction.

Exception handling: Invalid quantity or negative resulting stock → reject.

UC-05 Manage Suppliers

Main flow: Add/update supplier → validate required details → save.

Exception handling: Missing/invalid required data → reject.

UC-06 Create Purchase Order

Main flow: Select supplier → add items/quantities → calculate total → save order.

Exception handling: Inactive supplier or invalid item/quantity → reject.

UC-07 Receive Purchase

Main flow: Select purchase → confirm received quantities → validate → record receipt → increase stock.

Exception handling: Received quantity beyond allowed amount → reject.

UC-08 Record Sale

Main flow: Select products → enter quantities → validate stock → calculate total → complete sale → reduce stock.

Exception handling: Insufficient stock → reject and do not create sale.

UC-09 View Reports

Main flow: Select report/filter → retrieve and aggregate data → display results.

Exception handling: Invalid filter range → validation message.

UC-10 Search and Filter

Main flow: Enter search/filter → retrieve matches → display results/empty state.

Exception handling: No matches → show empty state.

9.4 Core Entity Analysis

Entity	Key Relationships
User / Role	A user has one role; the role controls permissions.
Category / Product	A category can contain multiple products; each product references a category.
Supplier / PurchaseOrder	A supplier can have multiple purchase orders; each purchase order references one supplier.
PurchaseOrder / PurchaseItem / Product	Each purchase contains one or more items; each item references a product and quantity.
Sale / SaleItem / Product	Each sale contains one or more items; each item references a product and quantity.
Product / InventoryTransaction	Inventory transactions record stock changes for products.

The final ER/UML diagrams should be added to the design documentation once the physical schema is frozen. Section 14 is the initial logical data model.

10. Functional Requirements

Requirements are grouped by system feature. Each is uniquely identified and written as mandatory, testable behavior.

10.1 Authentication & Role Management

Priority: High. Primary owner: Annamalai N.

Requirement ID	Requirement
FR-AUTH-001	The system shall allow a registered user to log in using valid credentials.
FR-AUTH-002	The system shall reject authentication when supplied credentials are invalid.
FR-AUTH-003	The system shall prevent a deactivated user from starting an authenticated session.
FR-AUTH-004	The system shall terminate the authenticated session when the user logs out.
FR-AUTH-005	The system shall associate each active user with an application role.
FR-AUTH-006	Only authorized administrators shall create, update, deactivate or reactivate user accounts.
FR-AUTH-007	The backend shall enforce role permissions for protected API operations.
FR-AUTH-008	The system shall return an access-denied response for operations outside the user's permission set.

10.2 Product & Category Management

Priority: High. Primary owner: Apurv Kumar Singh.

Requirement ID	Requirement
FR-PROD-001	The system shall allow authorized users to create a product with unique SKU, name, category, selling price and minimum stock level.
FR-PROD-002	The system shall reject creation/update of a product when its SKU duplicates an existing active SKU.
FR-PROD-003	The system shall allow authorized users to update product master data subject to validation.
FR-PROD-004	The system shall allow authorized users to create and maintain product categories.
FR-PROD-005	The system shall allow permitted users to search products by SKU or name.
FR-PROD-006	The system shall prevent physical deletion of products referenced by historical transactions; such products shall be deactivated.

10.3 Inventory Management

Priority: High. Primary owner: Apurv Kumar Singh.

Requirement ID	Requirement
FR-INV-001	The system shall display current available quantity for each active product.
FR-INV-002	The system shall allow permitted users to record a stock adjustment with product, quantity change, reason, user and timestamp.
FR-INV-003	The system shall reject an adjustment that would result in negative available quantity.
FR-INV-004	The system shall identify a product as low stock when available quantity is less than or equal to minimum stock.
FR-INV-005	The system shall record purchase, sale and adjustment events in inventory history.
FR-INV-006	The system shall provide inventory history filterable by product and date range.

10.4 Supplier & Purchase Management

Priority: High. Primary owner: Atharv Mittal.

Requirement ID	Requirement
FR-PUR-001	The system shall allow authorized users to create and update supplier records with required details.
FR-PUR-002	The system shall allow suppliers to be deactivated without deleting historical purchase records.
FR-PUR-003	The system shall allow purchase staff to create a purchase order for an active supplier.
FR-PUR-004	A purchase order shall contain at least one item with product, quantity and purchase price.
FR-PUR-005	The system shall calculate purchase order totals from item quantities and purchase prices.
FR-PUR-006	The system shall allow purchase staff to record received quantities against an existing purchase order.
FR-PUR-007	A successfully recorded receipt shall increase product inventory by the received quantity.
FR-PUR-008	The system shall prevent received quantity from exceeding ordered quantity unless an approved exception process exists.
FR-PUR-009	The system shall provide purchase history filterable by supplier and date range.

10.5 Sales Management

Priority: High. Primary owner: Avaneesh H D.

Requirement ID	Requirement
FR-SALE-001	The system shall allow authorized sales users to create a sale containing one or more items.
FR-SALE-002	The system shall validate each sale quantity as a positive integer.
FR-SALE-003	The system shall prevent a sale from being completed when requested quantity exceeds available stock.
FR-SALE-004	The system shall calculate the sale total from item quantities and selling prices.
FR-SALE-005	A completed sale shall decrease product inventory by the quantity sold.
FR-SALE-006	The system shall assign a unique transaction identifier to each completed sale.
FR-SALE-007	The system shall provide sales history filterable by transaction identifier and date range.

10.6 Reporting & Monitoring

Priority: High. Primary owner: Avaneesh H D.

Requirement ID	Requirement
FR-REP-001	The system shall provide an inventory report containing product, SKU, category, available quantity and stock status.
FR-REP-002	The system shall provide a low-stock report containing products at or below minimum stock.
FR-REP-003	The system shall provide a purchase report filterable by supplier and date range.
FR-REP-004	The system shall provide a sales report filterable by date range.
FR-REP-005	The system shall display report totals consistently with underlying stored transactions.

10.7 Common Validation & Search

Priority: Medium. Primary owner: Shared.

Requirement ID	Requirement
FR-COM-001	The system shall validate required fields before create/update operations.
FR-COM-002	The system shall display a clear validation message for each invalid input that prevents submission.
FR-COM-003	The system shall display a success/confirmation message after successful create, update, receipt, adjustment or sale operations.
FR-COM-004	The system shall display an explicit empty-state message when a valid search/filter returns no records.
FR-COM-005	The system shall preserve referential integrity when records are updated or deactivated.

11. Non-Functional Requirements

ID	Category	Measurable Requirement	Verification
NFR-001	Performance	At least 95% of normal authenticated API requests shall respond within 2 seconds under the defined test load.	Performance test
NFR-002	Availability	The application shall achieve at least 99% availability during the defined operating window, excluding planned maintenance.	Operational review
NFR-003	Usability	A trained user shall complete a basic product search and stock lookup within 3 primary screens.	Usability walkthrough

ID	Category	Measurable Requirement	Verification
NFR-004	Reliability	Retrying the same failed transaction request shall not create duplicate purchase or sale records.	Integration/negative test
NFR-005	Maintainability	Business functionality shall be separated into identifiable modules with automated tests for core services.	Code review
NFR-006	Testability	Core API services shall expose deterministic behavior suitable for automated unit/integration tests.	Test execution
NFR-007	Portability	The web client shall work on current stable Chrome/Chromium, Firefox and Edge during final testing.	Cross-browser test
NFR-008	Scalability	The schema shall support at least 10,000 products and 100,000 transaction/item records without schema redesign.	Data-volume test
NFR-009	Data Integrity	Purchase, sale and inventory updates shall be atomic so partial updates cannot leave inconsistent stock.	Database/integration test
NFR-010	Usability	Validation/error messages shall identify the field or operation needing correction.	UI test
NFR-011	Maintainability	Code shall follow agreed formatting, naming, modularity, Git review and static-analysis practices.	Code review/SonarQube
NFR-012	Auditability	Inventory-affecting events shall retain responsible user and timestamp where applicable.	Database inspection/test

12. Security Requirements

ID	Security Requirement	Verification
SEC-001	The system shall authenticate users before granting access to protected functions.	Authentication test
SEC-002	Passwords shall not be stored in plaintext and shall use secure password hashing.	Code/database review
SEC-003	Authorization shall be enforced at the server/API layer.	Authorization test
SEC-004	Client-side manipulation shall not allow access outside the user's permissions.	Negative API test
SEC-005	Credentials and authenticated session/token data shall be transmitted only over HTTPS in production.	Network/deployment inspection
SEC-006	The backend shall validate and sanitize client-supplied input before database operations.	Security tests
SEC-007	Secrets and database credentials shall not be exposed in client-side code.	Configuration/code review
SEC-008	Authentication failures shall not reveal whether a particular account exists.	Security test
SEC-009	Inactive accounts shall not authenticate successfully.	Authentication test
SEC-010	Database access shall use least-privilege application credentials.	Deployment review

13. Business / Domain Rules

ID	Business / Domain Rule
BR-001	Every active product shall have a unique SKU.
BR-002	Available stock quantity shall never be negative.
BR-003	Low stock means available quantity is less than or equal to minimum stock.
BR-004	Only active suppliers may be used for new purchase orders.
BR-005	A valid purchase order shall contain at least one item.
BR-006	Receiving a purchase increases inventory by the quantity actually received.
BR-007	A completed sale decreases inventory by the quantity sold.
BR-008	A sale cannot be completed when requested quantity exceeds available stock.
BR-009	Historical transactions remain valid when a product or supplier is deactivated.
BR-010	Only authorized roles may perform administrative, inventory-adjustment, purchasing or sales operations.
BR-011	Transaction totals shall be derived from stored line quantities and unit prices.
BR-012	Inventory changes shall be traceable to their source purchase, sale or adjustment event.

13.1 Permission Matrix

Function	Admin	Inventory	Purchase	Sales	Manager
User/Role Management	C/U/D	-	-	-	View
Product/Category	C/U/D*	C/U	View	View	View
Inventory View	Yes	Yes	Yes	Yes	Yes
Inventory Adjustment	Yes	Yes	-	-	Policy
Supplier Management	C/U/D*	View	C/U/D*	View	View
Purchase Orders	Yes	View	C/U/Receive	View	View
Sales	Yes	View	View	Create/View	View
Reports	All	Inventory	Purchase	Sales	All

* C = Create, U = Update, D = Deactivate/retire. Historical records shall not be physically deleted when references exist.

14. Database / Field Requirements

These are the initial logical field requirements. Physical column names, indexes and implementation-specific constraints may be refined during database design without changing the intended behavior.

14.1 User and Role Fields

Entity	Field	Type	Length/Range	Mandatory	Description
User	user_id	PK	Generated	Y	Unique user identifier
User	name	String	1–100	Y	Display/full name
User	email	String	Valid email	Y	Unique login/contact identifier
User	password_hash	String	Implementation-defined	Y	Secure password hash
User	role_id	FK	Valid role	Y	Assigned role

Entity	Field	Type	Length/Range	Mandatory	Description
User	status	Enum	ACTIVE/INACTIVE	Y	Account status
Role	role_id	PK	Generated	Y	Unique role ID
Role	role_name	String	1–50	Y	Role name

14.2 Product, Category and Inventory Fields

Entity	Field	Type	Length/Range	Mandatory	Description
Category	category_id	PK	Generated	Y	Unique category ID
Category	name	String	1–80	Y	Category name
Product	product_id	PK	Generated	Y	Unique product ID
Product	sku	String	1–40	Y	Unique stock keeping unit
Product	name	String	1–150	Y	Product name
Product	category_id	FK	Valid category	Y	Category reference
Product	selling_price	Decimal	>=0	Y	Unit selling price
Product	minimum_stock	Integer	>=0	Y	Low-stock threshold
Product	status	Enum	ACTIVE/INACTIVE	Y	Product status
Inventory	product_id	FK/PK	Valid product	Y	Referenced product
Inventory	available_qty	Integer	>=0	Y	Current stock
InventoryTransaction	transaction_id	PK	Generated	Y	Inventory event ID
InventoryTransaction	product_id	FK	Valid product	Y	Affected product
InventoryTransaction	change_qty	Integer	Signed	Y	Quantity change
InventoryTransaction	source_type	Enum	PURCHASE/SALE/ADJUSTMENT	Y	Origin
InventoryTransaction	source_id	Reference	Valid source	Y	Source transaction ID
InventoryTransaction	created_by	FK	Valid user	Y	Responsible user
InventoryTransaction	created_at	Timestamp	Generated	Y	Event timestamp

14.3 Supplier and Purchase Fields

Entity	Field	Type	Length/Range	Mandatory	Description
Supplier	supplier_id	PK	Generated	Y	Supplier ID
Supplier	name	String	1–150	Y	Supplier/business name
Supplier	contact_name	String	0–100	N	Primary contact
Supplier	phone	String	Valid phone	N	Contact number
Supplier	email	String	Valid email	N	Contact email
Supplier	address	String	0–250	N	Supplier address
Supplier	status	Enum	ACTIVE/INACTIVE	Y	Supplier status
PurchaseOrder	purchase_id	PK	Generated	Y	Purchase order ID
PurchaseOrder	supplier_id	FK	Valid supplier	Y	Supplier reference
PurchaseOrder	order_date	Date	Valid date	Y	Order date
PurchaseOrder	status	Enum	DRAFT/ORDERED/PARTIAL/RECEIVED/CANCELLED	Y	Purchase status
PurchaseOrder	total_amount	Decimal	>=0	Y	Derived order total
PurchaseItem	purchase_item_id	PK	Generated	Y	Line ID
PurchaseItem	purchase_id	FK	Valid purchase	Y	Parent purchase
PurchaseItem	product_id	FK	Valid product	Y	Product reference
PurchaseItem	ordered_qty	Integer	>0	Y	Ordered quantity
PurchaseItem	received_qty	Integer	0..ordered	Y	Received quantity
PurchaseItem	unit_cost	Decimal	>=0	Y	Purchase unit cost

14.4 Sales Fields

Entity	Field	Type	Length/Range	Mandatory	Description
Sale	sale_id	PK	Generated	Y	Sale identifier
Sale	transaction_no	String	Unique	Y	Human-readable transaction reference
Sale	sale_date	Timestamp	Generated	Y	Sale timestamp
Sale	created_by	FK	Valid user	Y	Sales user
Sale	total_amount	Decimal	>=0	Y	Derived sale total
SaleItem	sale_item_id	PK	Generated	Y	Line ID
SaleItem	sale_id	FK	Valid sale	Y	Parent sale
SaleItem	product_id	FK	Valid product	Y	Sold product

Entity	Field	Type	Length/Range	Mandatory	Description
SaleItem	quantity	Integer	>0	Y	Quantity sold
SaleItem	unit_price	Decimal	>=0	Y	Price at sale time
SaleItem	line_total	Decimal	>=0	Y	Quantity × unit price

14.5 Report Requirements

Report	Required Fields / Filters
Inventory Report	SKU, product, category, available quantity, minimum stock, stock status; category/status filters.
Low-Stock Report	SKU, product, available quantity, minimum stock, status; category filter.
Purchase Report	Purchase ID, supplier, order date, status, total; supplier/date filters.
Sales Report	Transaction number, sale date, total, created by; date filter.
Inventory History	Timestamp, product, change quantity, source type/source ID, user; product/date filters.

15. Glossary

Term	Definition
API	Application Programming Interface used for communication between software components.
CRUD	Create, Read, Update and Delete operations on data.
Django REST Framework	Python framework planned for implementing REST APIs.
Inventory	Current quantity of products available in the store.
Inventory Adjustment	Authorized stock quantity change recorded with a reason.
Low Stock	Condition where available quantity is less than or equal to minimum stock.
Purchase Order	Record representing goods ordered from a supplier.
Purchase Receipt	Record of goods actually received against a purchase order.
RBAC	Role-Based Access Control; permissions are assigned according to user roles.
REST	Representational State Transfer style used for web APIs.
SKU	Stock Keeping Unit; unique product identifier.
PostgreSQL	Relational database planned for persistent application data.
React	JavaScript library planned for the web frontend.
Sale	Recorded transaction in which one or more products are sold.
Supplier	Business/entity from which the store procures products.
TechZone Electronics	Fictional electronics retail shop used as the project case-study context.

16. Requirement Traceability Matrix

The RTM links selected requirements to planned architecture, design/use-case, code and test artifacts. Since implementation is not yet complete, code and test references are planned identifiers and shall be updated during development.

Req ID	Brief Description	Architecture	Design/UC	Planned Code	Test ID
FR-AUTH-001	Valid login	AUTH-MOD	UC-01	auth/	TC-AUTH-01
FR-AUTH-007	Backend authorization	AUTH-SEC	UC-02	permissions/	TC-SEC-03
FR-PROD-001	Create product	PRODUCT-MOD	UC-03	products/	TC-PROD-01
FR-PROD-002	Reject duplicate SKU	PRODUCT-MOD	UC-03	products/validators	TC-PROD-02
FR-INV-001	Display stock	INVENTORY-MOD	UC-04	inventory/	TC-INV-01
FR-INV-004	Low-stock identification	INVENTORY-MOD	UC-04/09	inventory/rules	TC-INV-04
FR-PUR-003	Create purchase order	PURCHASE-MOD	UC-06	purchases/	TC-PUR-03
FR-PUR-007	Increase stock on receipt	PURCHASE+INVENTORY	UC-07	purchases/receipt	TC-PUR-07
FR-SALE-003	Prevent sale above stock	SALES+INVENTORY	UC-08	sales/	TC-SALE-03

Req ID	Brief Description	Architecture	Design/UC	Planned Code	Test ID
FR-SALE-005	Decrease stock after sale	SALES+INVENTORY	UC-08	sales/checkout	TC-SALE-05
FR-REP-001	Inventory report	REPORT-MOD	UC-09	reports/	TC-REP-01
FR-REP-004	Sales report	REPORT-MOD	UC-09	reports/	TC-REP-04
FR-COM-001	Required-field validation	COMMON-MOD	All relevant	validators/	TC-COM-01
NFR-001	2-second API target	ARCH-PERF	System	API/services	TC-NFR-01
NFR-009	Atomic updates	DB-ARCH	UC-07/08	transaction services	TC-NFR-09
SEC-002	Password hashing	AUTH-SEC	UC-01	auth config	TC-SEC-02
SEC-004	Block unauthorized API access	AUTH-SEC	UC-02+	permissions/	TC-SEC-04
BR-003	Low-stock rule	INVENTORY-DOMAIN	UC-04/09	inventory rules	TC-BR-03
BR-008	Sale cannot exceed stock	SALES-DOMAIN	UC-08	sales rules	TC-BR-08

16.1 Requirement Ownership Coverage

Team Member	Primary Functional Area	Requirement Groups Owned
Annamalai N	Authentication & Role Management	FR-AUTH-001 to FR-AUTH-008; related SEC requirements
Apurv Kumar Singh	Product & Inventory Management	FR-PROD-001 to FR-PROD-006; FR-INV-001 to FR-INV-006
Atharv Mittal	Supplier & Purchase Management	FR-PUR-001 to FR-PUR-009
Avaneesh H D	Sales & Reporting Management	FR-SALE-001 to FR-SALE-007; FR-REP-001 to FR-REP-005

Shared requirements cover validation, common UI behavior, data integrity, testing, documentation and cross-module integration. Each member therefore owns a complete business functionality.

Conclusion

This SRS establishes the initial requirements baseline for the TechZone Electronics Inventory Management System. It defines scope, users, functions, interfaces, use cases, measurable quality requirements, security controls, business rules, logical data fields, glossary terms and traceability. Approved changes shall be reflected through the revision history and RTM.